

TrackMind

Business Analysis and Unit Economics

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1. General Context

The European railway market is governed by a complex regulatory framework that mandates a technical certification process for every train before it can operate on any national network. This process becomes significantly more complex when a vehicle needs to operate across multiple countries, as each member state maintains national rules that supplement the common European standards.

1.1 Technical Specifications for Interoperability (TSI)

TSIs are European regulations that define the technical requirements every railway component must meet to ensure interoperability across the networks of different member states. Each TSI covers a specific subsystem: the LOC&PAS TSI governs requirements for locomotives and passenger rolling stock and runs to over 300 pages of verifiable technical requirements.

1.2 The Cross-Border Certification Problem

When a manufacturer seeks authorisation to operate a vehicle on a foreign network, complying with the common TSI requirements is not sufficient: they must also verify conformity with the National Technical Rules (NTR) of the destination country. These rules:

- Exist in 24 different languages, one per EU member state
- Are additional to TSIs and are often untranslated
- May require national tests or standards that differ from European ones (e.g. France: NF F31-054 vs EN 14752 for door obstacle detection systems)
- Change frequently, requiring continuous monitoring by compliance engineers

The process of researching and consulting these regulations is entirely manual today: a specialised engineer must read the applicable TSI, locate the NTR of the target country in its original language, compare them requirement by requirement, and identify any conflicts. This process requires between 4 and 6 hours for each complex compliance query.

The problem in one sentence

A Spanish manufacturer seeking to extend the operating area of their train to the French network must cross-reference hundreds of technical requirements across documents in different languages, with no automated support. TrackMind automates this research, reducing query resolution time from hours to minutes.

1.3 The Authorisation Process: From Manufacturer to ERA

The path to final authorisation for a cross-border vehicle involves three sequential phases and three distinct players:

Phase	Who acts	What is produced
1. Document preparation	Manufacturer (internal engineers)	Technical file: the document demonstrating conformity with all applicable TSI and NTR requirements
2. Verification and certification	NoBo (Notified Body)	EC Verification Certificate: independent technical proof that the vehicle meets TSI requirements
3. Final authorisation	ERA (European Union Agency for Railways; for multi-country projects) or national NSA	Official authorisation to place the vehicle into service on the target network

The NoBo does not issue the final authorisation — it issues the EC certificate which is a necessary prerequisite. Final authorisation is granted by ERA for multi-country projects.

1.4 Market Scale: How Large Is the Opportunity?

The scale of the certification market that TrackMind targets can be quantified from public ERA data. Since the 4th Railway Package entered into force in June 2019, ERA has authorised 80,000 vehicles through the One-Stop Shop, with 83% of those authorisations covering multiple EU countries. ERA now handles approximately 22% of all single safety certificates issued in the EU and continues to take on a larger share as the transition period ends. Every multi-country vehicle authorisation in this pipeline requires a cross-border technical file, which is exactly where TSI-versus-NTR conflicts arise and where TrackMind operates.

The European rolling stock market provides the upstream context. It is valued at approximately EUR 120 billion, growing at around 3.5% annually, and is consolidating around five global manufacturers: Alstom, Siemens Mobility, Stadler, CAF, and CRRC. These five account for roughly 71-81% of global rolling stock market share. Below them sits a second tier of mid-size European manufacturers — Talgo, Skoda Transportation, Newag (acquired by CAF in 2024), PESA, Bozankaya — who are precisely the companies with cross-border ambitions but without dedicated internal compliance teams large enough to handle multi-country NTR research efficiently.

On the certification body side, more than 40 Notified Bodies are registered in the NANDO database and participate in the NB-Rail Coordination Group. The major ones — TUV SUD, Bureau Veritas, Lloyd's Register, Ricardo Certification, CERTIFER, Italcert, SGS, SINTEF, Arsenal Race — operate across Europe and compete on price and delivery speed. Each processes multiple certification projects simultaneously. TUV SUD alone certified 34 vehicle types to TSI requirements in 2024, with an average assessment time of 4.2 months per LOC&PAS structural integrity review.

The TAM for TrackMind in the beachhead segment (mid-size manufacturers, Phase 1) is deliberately narrow: there are roughly 15-30 European mid-size rolling stock manufacturers with active cross-border programmes. At EUR 12,000/year per client subscription, the Phase 1 addressable revenue ceiling is approximately EUR 180,000-360,000 ARR. The Phase 2 expansion to NoBos opens a materially larger and higher-value segment, with NoBo contracts priced at EUR 2,500/month given the significantly higher value generated for their business model. Large manufacturers represent a further long-term expansion opportunity beyond the three-year horizon of this plan.

1.5 Use Cases: When Does TrackMind Create Value?

TrackMind creates value at two distinct moments in the cross-border certification lifecycle. These two use cases differ in intensity, frequency, and the strength of the competitive differentiation TrackMind offers. Understanding the distinction is important for both product positioning and revenue model design.

Primary use case: pre-submission compliance verification

The primary and most commercially compelling use case is the final verification of the technical file before it is submitted to the NoBo. At this stage, the manufacturer's documentation is complete or near-complete, and the engineering team must systematically verify that every applicable TSI requirement is addressed and that no conflict exists with the National Technical Rules of the target country. This is exactly the workflow TrackMind is designed for: tri-source retrieval across the TSI, the target country's NTR, and the manufacturer's own technical specification, followed by structured conflict detection and confidence-scored output.

The competitive differentiation here is strongest and clearest. A general-purpose LLM cannot replicate this workflow: it does not have access to the manufacturer's proprietary technical file, it does not maintain an indexed and up-to-date corpus of NTR documents across 24 EU member states, and it cannot produce a structured output with source citations and an audit trail suitable for a safety-critical

regulatory process. This use case is the core value proposition of TrackMind and the primary justification for the subscription price.

A direct consequence of better-prepared documentation is that the NoBo assessor spends less time on clarification rounds and iteration cycles. Since NoBo fees are at least partially variable and based on time spent, manufacturers who use TrackMind to produce cleaner technical files benefit not only from reduced internal engineering hours but also from reduced NoBo costs — a compounding saving that strengthens the ROI argument.

Secondary use case: regulatory intelligence during design and engineering

Earlier in the project lifecycle — during the design and engineering phase, before the technical file is complete — TrackMind can serve as a regulatory intelligence tool. An engineer designing a door system for a train intended to operate in both Spain and France can query TrackMind to retrieve the applicable dimensional and safety requirements from both countries' NTR simultaneously, identifying potential conflicts at the design stage rather than discovering them during pre-submission review when corrections are far more costly.

The differentiation in this use case is real but more nuanced. A general-purpose LLM, given the right prompting, could retrieve some regulatory information on a specific question. However, TrackMind remains superior for three reasons: it retrieves from an indexed, maintained, and current NTR corpus rather than from potentially outdated training data; it cross-references multiple national rules simultaneously in a single structured query; and it produces source-cited output that an engineer can reference in project documentation. That said, this use case is best positioned as a valuable add-on rather than the primary sales argument — it increases the perceived value of the subscription and deepens product engagement, but it is the pre-submission workflow that justifies the cost.

Summary

TrackMind is sold and priced as a pre-submission compliance tool — this is where the ROI is clearest, the differentiation is strongest, and the willingness to pay is highest. The design-phase regulatory intelligence capability is a natural extension of the same underlying system that increases product stickiness and reduces churn, as engineers who rely on TrackMind throughout the project lifecycle are significantly less likely to cancel at renewal.

2. The Players in the Market

The European railway certification market involves four categories of players with very different roles, cost structures, and incentives.

2.1 Manufacturers (Rolling Stock Producers)

Manufacturers are the companies that design and build trains, trams, metros, and specialised railway vehicles. They are the direct applicants for authorisation and the primary payers of the entire certification process.

Three size segments can be identified, each with very different characteristics:

Characteristic	Large manufacturers	Mid-size manufacturers	Small manufacturers
Examples	Alstom, Siemens Mobility, CAF, Stadler, Hitachi Rail	Specialised producers of trams, light metro, regional rolling stock	Subsystem and component suppliers
Indicative revenue	Billions of EUR	EUR 100-500 million	Below EUR 100 million
Internal compliance team	Dedicated team, 10+ TSI specialists	1-3 engineers, partially dedicated	Absent or minimal
Cross-border projects per year	Many, simultaneous	Several, moderate frequency	Rare
SaaS sales cycle	12-24 months (centralised procurement)	3-6 months (direct decision)	Long but limited budget

Manufacturer cost structure in the certification process

A manufacturer incurs three categories of cost in the cross-border certification process:

- Internal engineers (semi-fixed cost): salaries of staff partially dedicated to TSI document research. For a senior specialised engineer in Western Europe: approximately EUR 100,000/year all-in, of which an estimated 30% is dedicated to document research activities = EUR 30,000/year per engineer.
- NoBo assessor time (variable cost): the NoBo bills by time spent. Better-prepared documentation reduces iteration cycles and therefore the variable portion of NoBo fees. Reference rate: EUR 265/hour (proxy: ERA official hourly rate, source: ERA Executive Director Decision, December 2023).
- Delay costs (opportunity cost): every month of delay in authorisation on a EUR 500M contract can cost EUR 100,000-500,000. Difficult to quantify but potentially the largest cost driver.

2.2 NoBos (Notified Bodies)

NoBos are private bodies accredited and notified to the European Commission that perform mandatory independent technical verification. Without their EC certificate, no vehicle obtains authorisation. Key examples: TUV SUD, Bureau Veritas, Lloyd's Register, Ricardo Certification, CERTIFER.

Although accredited by public institutions, NoBos are private commercial companies competing with each other on price and delivery speed.

NoBo economic structure

The economic structure of a NoBo mirrors that of a professional services firm — analogous to a law firm or consulting company:

Item	Description
Revenue	Hourly rate x billable hours per project x number of projects managed
Main cost	Personnel: TSI-specialised assessors are rare and expensive figures
Margin	Depends on how many hours can be billed relative to hours actually spent per project
Competitive lever	Speed of delivery and ability to manage more projects simultaneously with the same headcount

2.3 ERA (European Union Agency for Railways)

ERA is the European public agency headquartered in Valenciennes, France. It is not a potential customer — it is the system regulator. Its main roles:

- Issues final authorisations for vehicles intended to operate in more than one member state
- Maintains the public databases ERADIS and ERATV with all certifications and authorisations
- Publishes and updates TSIs and their application guides
- Coordinates the One-Stop Shop (OSS), the single portal for authorisation applications

ERA's official hourly rate for processing applications is EUR 265/hour (ERA Executive Director Decision, December 2023 — primary public source). This figure is used as a conservative proxy in sector cost estimations.

2.4 Competitive Landscape: Who Else Is in This Space?

The competitive landscape for TrackMind must be understood across three distinct layers: the regulatory infrastructure that sits around the problem, the adjacent digital tools already present in the railway sector, and the general AI tools that could theoretically be applied to it. None of these constitutes a direct competitor today. The gap that TrackMind fills — AI-powered cross-document compliance reasoning at the TSI/NTR intersection — is unoccupied.

Layer 1: Regulatory infrastructure (not competitors — context)

ERA's One-Stop Shop (OSS) is the submission portal for completed vehicle authorisation applications. It is where the manufacturer's finished technical file goes after all TSI/NTR research is done. TrackMind operates upstream of the OSS, in the preparation phase. There is no overlap: the OSS is a process management tool, not a document intelligence tool. RailNetEurope (RNE) and the TAF/TAP TSI ecosystem — including players like OLTIS Group, RailData.coop, and the Wavestone/SNCF telematics platforms — address data exchange standards between railway undertakings and infrastructure managers for trains already in operation. Their TSI is about message formats for freight tracking and passenger ticketing. TrackMind's TSI is about technical safety specifications for vehicles being built. These are different regulations, different customers, and a different moment in the railway lifecycle entirely.

Layer 2: Adjacent digital tools in the railway sector

Every significant AI investment in European rail targets physical and operational problems: track inspection, predictive maintenance, and autonomous driving. Siemens Railigent X is an AI platform for operational analytics and predictive maintenance. Alstom HealthHub does fleet health monitoring. Network Rail deploys AI-powered track inspection that delivers over £20 million in annual productivity savings. CrossTech's Hubble system processes forward-facing camera video to detect vegetation and track defects across 8,000+ miles monthly. None of these touches the document intelligence layer for rolling stock certification. The reason is structural: physical AI for operations is a well-understood problem domain with large, established customers (national rail operators). Document AI for regulatory compliance in railway certification is a niche requiring deep EU regulatory knowledge and relationships

with a relatively small number of NoBos and mid-size manufacturers — it is too specialised for platform companies and too regulated for generic AI startups.

TUV SUD, Bureau Veritas, and other large NoBos do have internal digital workflow tools. TUV SUD's digital case management system is referenced in ERA's 2024 guidance on efficient NoBo processes, where it reduced administrative overhead by 31%. These are case management and scheduling tools, not compliance reasoning systems. They tell you where a certification project is in the process; they do not tell you whether a Spanish EMU's door detection system at 30mm satisfies the French NTR. The distinction matters for the competitive argument: those tools manage workflow around the document problem. TrackMind solves the document problem itself.

Layer 3: General AI tools

Claude, GPT-4o, and Gemini can read a TSI article if you paste it in. They cannot retrieve the right TSI article, find the relevant NTR for the target country in its original language, cross-reference both against the manufacturer's technical spec, assess primacy under Directive 2016/797, and produce a confidence-scored output with a full audit trail — all in one structured workflow. The document intelligence layer TrackMind provides requires a curated corpus, structured retrieval across three independent document collections, and a domain-specific reasoning system prompt. The general LLM is a component of that system, not a substitute for it. Document intelligence tools from adjacent domains — Trunk Tools (\$70M Series B, US construction document Q&A) and Civils.ai (UK construction search) — confirm that the RAG-over-regulatory-documents pattern is commercially validated, while also confirming that nobody has applied it to EU railway certification.

A further structural limitation of general-purpose AI tools is regulatory currency: their knowledge of TSI requirements and national rules is frozen at a training data cutoff. When ERA updates a TSI or a national authority revises an NTR, the general LLM remains unaware. TrackMind operates on a live, maintained corpus with a contractual SLA on regulatory currency (see Section 10) — a prerequisite for professional use in a regulated context that no general-purpose tool can match.

Competitive landscape summary

Company / Tool	What they do	Why not a direct competitor	Threat level
ERA One-Stop Shop	Submission portal for vehicle authorisations. Process management, not document intelligence.	Receives the finished technical file. TrackMind helps produce it. Entirely upstream.	None
RNE / OLTIS Group / TAF-TAP platforms	Data exchange standards for trains already running: freight tracking, ticketing, capacity planning.	Different regulation (TAF/TAP TSI vs LOC&PAS TSI), different customer (operators vs manufacturers), different lifecycle stage (operations vs certification).	None
Siemens Railigent X / Alstom HealthHub	Predictive maintenance, fleet health monitoring, operational analytics.	Physical/operational AI for running trains. No regulatory document intelligence. No overlap.	None
NoBo internal workflow tools (e.g. TUV SUD case management)	Project scheduling, document tracking, test witness coordination. Reduced NoBo admin	Manages workflow around the document problem. TrackMind solves the document	Low — potential future integration partner

Company / Tool	What they do	Why not a direct competitor	Threat level
	overhead by 31% (ERA 2024 guidance).	problem. Complementary, not competing.	
Trunk Tools / Civils.ai	AI Q&A and search over construction documents (drawings, specs, submittals). US and UK markets only.	Different domain (construction vs railway), different regulation, no EU market presence. Validates the RAG-over-documents pattern commercially.	None
General LLMs (Claude, GPT-4o, Gemini)	General-purpose text reasoning. Can read a TSI article if pasted in. No structured retrieval, no NTR corpus, no confidence scoring, no audit trail.	A component of TrackMind, not a substitute for it. The LLM is the reasoning engine; the RAG pipeline, NTR corpus, and workflow are what make it a product.	Medium — mitigated by domain depth and workflow integration

2.5 The AI Platform Threat: Why Can't Anthropic or OpenAI Just Build This?

Anthropic, OpenAI, and Google are infrastructure companies. Their product is the model and the API — they sell capability, not solutions. They have no incentive to build vertical applications in niche regulated markets: a TSI compliance tool for European railway certification would serve at most a few hundred organisations globally and would require Anthropic to maintain an active corpus of EU regulatory documents, stay current on TSI revisions, understand the NoBo accreditation structure, and navigate EU data residency rules for a customer segment whose procurement cycles run 12-24 months. None of this is in their product strategy.

The more realistic version of this threat is not that Anthropic builds TrackMind — it is that a well-funded horizontal legal or regulatory AI company expands into railway compliance. Harvey AI (legal document intelligence, \$300M raised as of 2024), Ironclad (contract intelligence), or a future entrant targeting EU regulatory domains could in principle replicate the RAG pipeline. The question is whether they would, and on what timeline. Three structural factors slow them down.

First: domain knowledge requires time, not just money.

TSI compliance is not generic legal document retrieval. The system must understand the relationship between EU directives and national derogations, the primacy rule under Directive 2016/797, the role of NB-Rail Recommendations for Use as interpretive consensus, the difference between NoBo scope (TSI) and DeBo scope (national rules), and the specific document structures of different member states' NNTR registers — many of which are not in English. A horizontal legal AI company entering this space would need 12-18 months of domain build before they could produce a credible product. TrackMind has a head start on that clock.

Second: the EU AI Act creates a compliance cost that late entrants must pay upfront.

TrackMind is classified as Limited Risk under the EU AI Act — not high-risk — because it is a preparatory tool that supports human decision-making without replacing it, and has no interface with safety-critical infrastructure (see Section 11). However, operating in a regulated safety-critical context requires a compliance posture that goes beyond the legal minimum: confidence-gated human review, a full audit trail, and versioned source citations are non-negotiable for any credible NoBo or manufacturer. TrackMind is built with these from day one. A new entrant must build this infrastructure from scratch, which represents a meaningful time and cost barrier to credible market entry.

Third: the data flywheel compounds with every approved resolution.

Every compliance query that passes through TrackMind and receives human approval becomes a labelled example of a correct TSI/NTR resolution. After 12-18 months of active client use, TrackMind will have a proprietary dataset of resolved cross-border compliance conflicts that no new entrant can replicate without starting over. This is the same flywheel dynamic that makes clinical decision support companies defensible against general medical AI: the labelled, domain-specific resolution history is the moat, not the underlying model.

The honest risk assessment.

The most credible near-term threat is not Anthropic or OpenAI entering the space directly. It is that AI model capabilities improve fast enough that a non-specialist — a compliance engineer with no coding background — can approximate TrackMind's output by manually feeding documents into a frontier model via a consumer interface. Models like Claude Projects and ChatGPT with file upload already allow this in a limited way. The defence is twofold: structured tri-source retrieval across large document corpora cannot be replicated by manual copy-paste at production scale, and the audit trail, confidence scoring, and human approval workflow that regulated clients require cannot be delivered by a consumer AI chat product.

AI Company / Platform	Realistic scenario	Why they are slowed down	Threat level
Anthropic / OpenAI / Google (model providers)	Build better models that power TrackMind. Will not build vertical railway compliance tools — wrong TAM, wrong customer, wrong geography focus.	Infrastructure companies. EU railway certification is too niche, too regulated, and too relationship-dependent for their product strategy.	Low (direct); Medium (indirect)
Harvey AI / horizontal legal AI platforms	Could expand into EU regulatory compliance verticals. Currently focused on legal contracts and case law — not technical safety standards.	12-18 months of domain build required. AI Act compliance infrastructure must be built from scratch. No existing TSI corpus or NoBo relationships.	Medium — realistic in 2-3 years
Frontier models with native file/project management (Claude Projects, ChatGPT Enterprise)	Engineers manually upload TSI and NTR documents and query the model directly. A rough approximation of TrackMind at zero cost.	No structured tri-source retrieval at scale. No confidence scoring. No audit trail. No multi-tenant document isolation. Consumer products cannot meet the compliance posture required for professional use in a regulated certification context.	Medium-High — the most credible near-term substitution risk

3. Choosing the Initial Target Customer

ERA is the regulator, not a market player. It does not purchase compliance software — it regulates it. We exclude it by definition.

3.1 Why We Exclude NoBos (at least initially)

NoBos (notified bodies) are in theory the most direct beneficiaries of a tool that accelerates document work: they would spend fewer hours per project, could handle more clients, and improve their margins. However, three barriers make NoBos the wrong customer for the MVP:

- **Regulatory barrier:** an accredited NoBo cannot adopt an unvalidated AI tool in its certification process without a formal internal validation cycle, with a 12-18 month sales cycle for enterprise customers of this type.
- **Legal liability:** the NoBo signs every EC certificate with its accreditation. Introducing an external tool into the process requires approval at very senior levels of the organisation and formal validation.
- **Reputational risk:** an AI system error on a compliance query in a safety-critical context could jeopardise the NoBo's accreditation. Structural resistance to adoption is high.

Conclusion: NoBos are the right customer for Phase 2, after TrackMind has built a verifiable track record with manufacturers. Not for the MVP.

3.2 Why We Exclude Large Manufacturers

Large manufacturers have the financial capacity to pay, but present three problems for an early-stage startup:

- **Very long sales cycle:** selling to Alstom or Siemens requires navigating centralised procurement, legal, IT security, and internal compliance — an estimated 12-24 months just to close the contract.
- **Asymmetric bargaining power:** a startup with no track record has no negotiating leverage with a multi-billion-euro company. The risk is accepting unsustainable terms or costly customisations.
- **Already structured internal teams:** large manufacturers have already invested in dedicated TSI compliance processes and personnel. The pain point exists but resistance to change is high.

3.3 Why Mid-Size Manufacturers Are the Ideal Customer

The mid-size manufacturer has the optimal profile to be TrackMind's first customer for three converging reasons:

Criterion	Assessment	Rationale
Adoption speed	High	The decision to adopt an internal tool rests with the technical director or engineering manager — no regulatory approval or centralised procurement required
Willingness to pay	High	Budget sufficient for a SaaS licence without board approval. ROI is immediately calculable on savings from reduced engineer time and NoBo iterations
Pain point intensity	Very high	Has cross-border projects but lacks internal resources to efficiently cover NTR in foreign languages. Document research is time-consuming and error-prone

Criterion	Assessment	Rationale
Sales cycle	3-6 months	Direct and accessible counterpart. Decision made by the person who lives the problem every day
Adoption risk	Low	TrackMind is an internal preparation tool — if it errs, the NoBo corrects it during review. The manufacturer signs nothing legally binding based on AI output

The go-to-market roadmap

Phase 1 (MVP, first 18 months): customer = mid-size manufacturer. Product positioned as a compliance preparation assistant. Build track record, accumulate the data flywheel, demonstrate accuracy and reliability on real cases. Phase 2 (after 18 months): customer = NoBos. TrackMind now has verifiable use cases and credible references. The NoBo can internally justify adoption with real accuracy data. NoBo contracts are priced at EUR 2,500/month reflecting the significantly higher value generated. Large manufacturers represent a potential further expansion beyond the three-year horizon of this plan.

4. The Mid-Size Manufacturer Cost Equation

4.1 The Three Cost Items in the Certification Process

A mid-size manufacturer incurs three categories of cost in the document research and verification process for cross-border projects. TrackMind directly impacts the first two, while the third is acknowledged but not quantified due to its complexity.

Cost item	Nature	TrackMind impact
A. Internal engineers (document research allocation)	Semi-fixed — engineer salary exists regardless of project activity	Reduced: hours spent on document research drop significantly per query
B. NoBo assessor time (variable component only)	Partially variable — NoBo bills by time spent; better documentation reduces iteration cycles	Partially reduced: cleaner technical files mean fewer clarification rounds with the NoBo
C. Delay costs	Opportunity cost — every month of authorisation delay on large contracts can cost EUR 100,000-500,000	Indirectly reduced: faster preparation lowers delay risk. Not quantified due to high variability.

4.2 How TrackMind Impacts Each Cost Item

Cost item A — Internal engineers

Without TrackMind, an engineer spends approximately 4 hours per compliance query: 45 minutes searching the TSI, 30 minutes locating the relevant NTR on the ERA database, 45 minutes checking the NB-Rail RFU database for prior resolutions, and 2 hours drafting the technical opinion.

With TrackMind, the AI retrieves and cross-references all three document layers simultaneously and drafts a structured response. The engineer reviews and approves in approximately 30 minutes (20 minutes AI processing + 10 minutes human review).

Parameter	Value	Source / Note
Queries per project	100	Midpoint of 50-200 range — original project proposal. Declared assumption.
Hours per query without TrackMind	4 hours	Breakdown: 45 min searching TSI for applicable requirements + 30 min locating NTR on ERA ERADIS + 45 min checking NB-Rail RFU database for prior resolutions + 2 hours drafting technical opinion.
Hours per query with TrackMind	0.5 hours	20 min AI + 10 min review
Engineer hourly cost	EUR 57/hour	EUR 100,000 all-in annual cost / 1,750 working hours. Declared assumption.
Cost without TrackMind (item A)	EUR 22,800	100 x 4h x EUR 57
Cost with TrackMind (item A)	EUR 2,850	100 x 0.5h x EUR 57

Parameter	Value	Source / Note
Saving on item A	EUR 19,950 per project	87.5% reduction in engineer time

Cost item B — NoBo assessor time (variable component)

Parameter	Value	Source / Note
NoBo days per query	2 days	Midpoint of 1-3 day range. Declared assumption.
NoBo hours per query	16 hours	2 days x 8 hours
Variable portion of NoBo time	30%	Conservative assumption — declared, not verified from primary source
Variable hours per query	4.8 hours	16h x 30%
NoBo hourly rate	EUR 265/hour	ERA Executive Director Decision, December 2023 — primary source (conservative proxy)
Variable NoBo cost per query without TrackMind	EUR 1,272	4.8h x EUR 265
Reduction with TrackMind	87.5%	Proportional to engineer time reduction
Variable NoBo cost per query with TrackMind	EUR 159	EUR 1,272 x 12.5%
Saving on item B (100 queries)	EUR 111,300 per project	100 x (EUR 1,272 - EUR 159)

4.3 Total Estimated Saving per Cross-Border Project

Cost item	Without TrackMind	With TrackMind	Saving per project
A — Internal engineers	EUR 22,800	EUR 2,850	EUR 19,950
B — NoBo variable component	EUR 127,200	EUR 15,900	EUR 111,300
C — Delay costs	Not quantified	Reduced	Not quantified
Total (A + B)	EUR 150,000	EUR 18,750	EUR 131,250

Key finding

The estimated total saving per cross-border project is approximately EUR 131,000. Even in a conservative reading — where the NoBo variable component saving is discounted entirely — the saving on internal engineer time alone (EUR 19,950) already exceeds the annual subscription cost of EUR 12,000 on a single project.

4.4 Methodological Limitations — A Frank Assessment

The cost estimates presented in this section are built on a chain of assumptions, several of which cannot be verified from public primary sources. We present this analysis transparently rather than concealing its limitations.

- The number of cross-language queries per project (100) is a midpoint estimate. It has not been verified against ERA data.
- The engineer hourly cost (EUR 57/hour) is derived from a EUR 100,000 all-in annual salary assumption. Actual salaries vary significantly by country and employer.
- The 30% variable component of NoBo time is a conservative assumption with no primary source backing.
- The NoBo hourly rate of EUR 265/hour is the ERA official rate used as a conservative proxy. Commercial NoBo rates are likely higher.
- The assumption that TrackMind reduces engineer time by 87.5% has not been validated through a working prototype at scale.
- The 4-hour baseline per query is derived from the following step-by-step workflow: 45 min searching the TSI, 30 min locating the NTR on ERA ERADIS, 45 min checking NB-Rail RFU for prior resolutions, 2 hours drafting the technical opinion.

Despite these limitations, we proceed with this analysis because even under the most conservative reading — using only the engineer time saving and excluding the NoBo component entirely — the estimated saving per project (EUR 19,950) already justifies the annual subscription cost on a single project.

5. Unit Economics

This section presents two distinct unit economics analyses for TrackMind. The two analyses are kept deliberately separate because the technology choices, cost structures, and assumptions differ significantly between a zero-cost academic demo and a production-grade SaaS platform.

The academic prototype was built using free and near-zero-cost tools — a deliberate choice for a university project where minimising spend is a priority and production-grade infrastructure is unnecessary. In contrast, a real commercial product would require paid infrastructure: a dedicated VM to run the embedding model without cold-start latency, a production hosting environment for the API backend, and an audit log database that supports concurrent multi-tenant access. These infrastructure costs are modest in absolute terms, but they are real and must be accounted for in any honest business model. The two analyses that follow make this distinction explicit.

5.1 Unit Economics — Academic Prototype

The academic prototype runs entirely on free or open-source tools hosted locally. The only real cost is the LLM inference API call — everything else runs on the developer's machine at zero marginal cost.

Component	Technology used	Cost per query	Note
LLM inference	Claude Sonnet 4.5 API	EUR 0.033	\$3/\$15 per MTok (Anthropic official pricing, May 2026). 7,000 input + 1,000 output tokens per query. Identical pricing to production model.
Embedding model	BAAI/bge-m3 local (HuggingFace)	EUR 0.000	Open-source model running locally via HuggingFace. Zero API cost. Same model used in production architecture.
Vector database	ChromaDB local (PersistentClient)	EUR 0.000	Local disk storage. No cloud hosting needed for a single-demo context.
PDF parsing	PyMuPDF (local, open-source)	EUR 0.000	Open-source library. Handles PDF chunking and metadata tagging locally at zero cost.
Frontend hosting	Streamlit Community Cloud	EUR 0.000	Free tier for academic and open projects.
Total cost per query	All components	EUR 0.033	100% driven by LLM inference. A 100-query demo costs EUR 3.30 in total.

The prototype costs EUR 0.033 per query — entirely driven by the LLM API call. All other components run locally at zero cost. The production architecture adds fixed infrastructure costs (VM hosting, API backend) but the per-query variable cost remains identical at EUR 0.033. The gap between the two models is not in the cost of intelligence — it is in the cost of making that intelligence reliable, scalable, and multi-tenant.

5.2 Unit Economics — Production Product

This section models the cost structure of TrackMind as a commercially launched SaaS product. All costs are expressed on a per-client, per-month basis to allow direct comparison with the subscription pricing model.

Key assumption: TrackMind serves 5 active clients simultaneously in Year 1. This is the baseline used to allocate all fixed costs across the client base.

5.2.1 Silicon Costs

Variable cost — LLM Inference (Claude Sonnet 4.5)

Model choice: Claude Sonnet 4.5 at \$3.00 per million input tokens and \$15.00 per million output tokens (source: Anthropic official pricing, May 2026). We chose Claude Sonnet over GPT-4o or Gemini Pro because it offers the best combination of long-context reasoning quality and cost for technical multilingual documents at production scale.

Token type	Tokens per query	Price per million tokens	Cost per query
Input tokens	7,000	\$3.00 / MTok	EUR 0.019
Output tokens	1,000	\$15.00 / MTok	EUR 0.014
Total LLM cost per query			EUR 0.033
Monthly LLM cost per client (100 queries)			EUR 3.30

Fixed costs — Embedding model (bge-m3 on GCP Compute Engine e2-standard-2, EUR 46/month): BAAI/bge-m3 runs locally on a dedicated VM, eliminating any per-query API cost for embeddings. It outperforms OpenAI text-embedding-3-small on European language pairs. Cost per client: EUR 46 / 5 clients = EUR 9.20 per client per month.

Fixed cost — Vector database (ChromaDB self-hosted on GCP VM, EUR 0/month additional): ChromaDB runs self-hosted on the same GCP VM as the bge-m3 embedding model, adding zero marginal infrastructure cost. Client document collections are isolated at the database level. Cost per client: EUR 0.

Fixed cost — API backend hosting (GCP Cloud Run, EUR 10/month estimated): Cloud Run scales to zero when idle, making it cost-effective for an early-stage product. Cost per client: EUR 10 / 5 clients = EUR 2.00 per client per month.

Fixed cost — Audit log database (Supabase free tier, EUR 0/month): provides managed PostgreSQL with a free tier that comfortably supports the audit log volume generated by 5 clients. Cost per client: EUR 0.

Component	Technology	Monthly cost (total)	Monthly cost per client	Nature
LLM inference	Claude Sonnet 4.5	Variable	EUR 3.30	Variable
Embedding model	bge-m3 on GCP e2-standard-2	EUR 46.00	EUR 9.20	Fixed
Vector database	ChromaDB self-hosted on GCP VM	EUR 0.00 (on existing VM)	EUR 0.00	Fixed
API backend hosting	GCP Cloud Run	EUR 10.00	EUR 2.00	Fixed
Audit log database	Supabase free tier	EUR 0.00	EUR 0.00	Fixed
Total silicon cost per client/month			EUR 14.50	

5.2.2 Carbon Costs

Carbon costs cover the human personnel required to build, maintain, and support the platform. For a SaaS AI startup at early stage, this is by far the dominant cost category.

In Year 1 (up to approximately 10 clients), TrackMind requires a single technical hire: a full-stack AI engineer who covers both the backend AI pipeline and the frontend product. A second engineer will be added once revenue from client growth makes it financially sustainable, typically around 15-20 active clients in Year 3, fully funded by the expanded revenue base.

Salary benchmark — Portugal (startup based at Nova SBE, Lisbon). Employer social security contributions in Portugal are 23.75% of gross salary (source: Boundless European Employment Costs Report 2025).

Role	Gross annual salary	Employer cost (+23.75%)	Source / Note
Full-stack AI engineer (single hire, early stage)	EUR 52,500	EUR 65,000	Glassdoor, Levels.fyi, SalaryExpert — AI/full-stack engineer Lisbon, May 2026.
Total employer cost (Year 1)	EUR 52,500/year	EUR 65,000/year	Single hire covering full stack. Second engineer added at 15-20 clients (Year 3).

The total annual technical personnel cost (EUR 65,000) is divided equally across the 5 clients assumed in Year 1: $\text{EUR } 65,000 / 12 \text{ months} / 5 \text{ clients} = \text{EUR } 1,083 \text{ per client per month}$.

5.2.3 Total Cost per Client per Month

Cost category	Component	Monthly cost per client
Silicon	LLM inference (100 queries x EUR 0.033)	EUR 3.30
Silicon	Embedding VM pro-rata (EUR 46 / 5)	EUR 9.20
Silicon	Vector DB (ChromaDB self-hosted, on existing VM)	EUR 0.00
Silicon	Backend hosting pro-rata (EUR 10 / 5)	EUR 2.00
Silicon	Audit log (Supabase free tier)	EUR 0.00
Silicon subtotal		EUR 14.50
Carbon	Full-stack AI engineer (EUR 65,000 / 12 / 5)	EUR 1,083.00
Carbon subtotal		EUR 1,083.00
TOTAL COST PER CLIENT PER MONTH		EUR 1,097.50

Key insight: silicon vs carbon

Silicon costs (EUR 14.50/month) represent only 1.3% of the total cost per client. Carbon costs (EUR 1,083/month) represent 98.7%. This is the defining economic characteristic of an early-stage AI SaaS startup: the marginal cost of serving an additional query is negligible, but the fixed cost of

maintaining the engineering team is substantial. The subscription price must be set primarily to cover the carbon cost, not the silicon cost. The fully-loaded gross margin improves significantly as the client base grows, because the carbon cost is semi-fixed.

5.2.4 How Costs Scale with Client Growth

Number of clients	Total silicon cost/month	Total carbon cost/month	Total cost/month	Cost per client/month
5 (Year 1 baseline)	EUR 96	EUR 5,417	EUR 5,490	EUR 1,098
10	EUR 191	EUR 5,417	EUR 5,585	EUR 559
20	EUR 382	EUR 10,834	EUR 11,216	EUR 561
50	EUR 955	EUR 10,834*	EUR 11,789	EUR 236

* A second engineer is added at 20 clients, increasing the monthly carbon cost to EUR 10,834 (2 x EUR 65,000/year / 12 months). The figures for 20 and 50 clients already reflect this. At 50+ clients, further headcount growth may be required and should be modelled in a detailed Year 3+ plan.

6. Revenue Model and Pricing

6.1 Pricing Model

TrackMind is offered as a flat monthly subscription with no setup fee and unlimited queries. The pricing structure is deliberately simple: one price, one plan, no usage caps, no onboarding barrier. The rationale for each design choice is as follows.

- No setup fee: document ingestion and initial indexing cost TrackMind less than EUR 1 per project in compute.
- Unlimited queries: a compliance engineer who must self-limit their queries to avoid overage charges will use the tool less, derive less value, and be more likely to cancel. Unlimited usage encourages full adoption and builds the data flywheel that strengthens the product over time.
- Flat monthly subscription: a predictable monthly cost is easy for the client to budget, easy to justify internally, and does not require board-level approval at EUR 1,000/month. It also aligns with TrackMind's strategic objective of building long-term client relationships rather than maximising per-project revenue.

Pricing component	Detail
Monthly subscription price	EUR 1,000 / month
Setup fee	None
Query limits	Unlimited
Contract structure	Monthly rolling.
Annual contract value (ACV)	EUR 12,000 / year

6.2 Customer ROI Analysis

The following ROI calculation is deliberately doubly conservative. It assumes (1) the manufacturer has only one cross-border project per year — the minimum plausible scenario — and (2) it excludes entirely the secondary saving on NoBo assessor time (cost item B), which is real but rests on less certain assumptions.

Item	Amount
Annual saving on engineer time (1 project, 100 queries, 4h to 0.5h per query, EUR 57/h)	EUR 19,950
Annual TrackMind subscription cost	EUR 12,000
Net annual saving for the client	EUR 7,950
ROI	66%
Payback period	~7.2 months

This estimate is doubly conservative. First, it assumes only one cross-border project per year — even one additional project doubles the gross saving to EUR 39,900, raising the net saving to EUR 27,900 and the ROI to 233%. Second, it excludes the secondary saving on NoBo assessor time (cost item B), which — if included under the intermediate scenario — would increase the total saving per project by over EUR 100,000.

6.3 Margin Analysis

At EUR 1,000/month per client and a total cost of EUR 1,097.50/month per client, TrackMind operates at a structural per-client loss of EUR 97.50/month with 5 clients. Break-even at steady state is reached at 6 clients.

Number of clients	Monthly revenue	Monthly total cost	Monthly margin
5 (Year 1)	EUR 5,000	EUR 5,490	-EUR 490
6 (break-even)	EUR 6,000	EUR 5,532	+EUR 468
10	EUR 10,000	EUR 5,585	+EUR 4,392

The initial loss of EUR 490/month at 5 clients disappears entirely at 6 clients and converts into a substantial positive margin as the client base grows. The path to profitability does not require a price increase — it requires client growth.

7. Go-to-Market Roadmap

TrackMind follows a two-phase go-to-market strategy within the three-year horizon of this plan. Each phase builds the conditions that make the next phase accessible: mid-size manufacturers generate the cash flow and accuracy track record that open the door to NoBos. Large manufacturers represent a potential long-term expansion beyond Year 3 and are not included in the current financial projections.

One important constraint shapes the entire roadmap: the sales cycle for mid-size manufacturers is 4-6 months. This means TrackMind will have no revenue in H1 of Year 1, even if sales conversations begin on day 1.

Phase	Timing	Customer focus	Key events and milestones
Phase 1	Year 1 — Beachhead	Mid-size manufacturers only	No revenue in H1 (sales cycle 4-6 months). First contracts signed in Q3. Target 5 clients by year end. Build accuracy track record and data flywheel. Begin informal relationship-building with NoBos — no selling, only listening and learning their validation requirements.
Phase 2	Year 2 — Expand	Mid-size mfrs + NoBo sales cycle begins	Grow to 10-12 mid-size manufacturer clients. Break-even crossed once client base exceeds 6. Launch formal NoBo sales cycle — no NoBo revenue expected in Year 2. Introduce annual prepay option for existing clients.
Phase 3	Year 3 — Scale	Mid-size manufacturers + NoBos	Target 15 mid-size manufacturer clients. First NoBo contracts close. Hire second engineer — fully funded by Year 3 revenue. Total client base: ~17 (15 mid-size manufacturers + 2 NoBos).

Note on NoBo timing: the formal sales cycle begins in Year 2 — not Year 1. A NoBo that sees a startup with 5 clients and 12 months of history has every reason to decline. A NoBo that sees 10-12 clients, a growing accuracy track record, and a team that has already spent a year understanding their validation requirements has a credible reason to engage. No NoBo revenue should be expected before Year 3.

8. Financial Projections

Note on scope and simplification

These financial projections are intentionally simplified. They are designed to provide a high-level overview of the business model and validate its structural soundness — not to serve as an analytical forecast. Several real cost categories are not modelled here, including sales and marketing expenditure, legal and administrative costs, DevOps support, and customer success resources. The purpose of this section is to demonstrate that the unit economics are sound, that the path to profitability is clear, and that the required seed investment is modest.

Item	Year 1	Year 2	Year 3
Active clients (end of year)	5 mid-size mfrs	12 mid-size mfrs	15 mid-size mfrs + 2 NoBos
Annual revenue	EUR 21,000	EUR 102,000	EUR 240,000
Total costs (personnel + silicon)	EUR 65,874	EUR 67,124	EUR 135,124
Operating margin	-EUR 44,874	+EUR 34,876	+EUR 104,876
ARR (end of year run rate)	EUR 60,000	EUR 144,000	EUR 240,000

Understanding the Year 1 loss

The Year 1 operating loss of EUR 44,874 is not a signal of a broken business model. It is a timing artefact driven by two predictable factors: the engineer salary begins from day 1 as a fixed cost, while revenue arrives only in month 4-6 at the earliest. The required runway is EUR 44,000-48,000 — a modest seed investment for a B2B SaaS startup. Break-even is reached in Year 2 as soon as the client base exceeds 6 active subscriptions. From that point, every additional client is pure operating leverage on a cost base that barely grows.

9. Data Protection and Security

TrackMind handles some of the most commercially sensitive documents a rolling stock manufacturer produces: technical files for vehicles under development, proprietary engineering specifications, and internal compliance assessments. Data protection is therefore not a compliance checkbox — it is a core product requirement and a meaningful competitive differentiator against general-purpose LLM alternatives.

9.1 Data Isolation Architecture

Every client's documents are stored in a dedicated, isolated ChromaDB collection on TrackMind's own GCP infrastructure. No client document ever resides in the same collection as another client's data. Retrieval queries are scoped exclusively to the requesting client's collection — it is architecturally impossible for a query from Manufacturer A to retrieve chunks from Manufacturer B's technical file.

Critically, all vector storage runs self-hosted on TrackMind's controlled infrastructure. Client documents are never transmitted to a third-party provider for indexing or storage. This means there is no third-party data processing agreement required for the vector database layer.

9.2 Data Ownership and Usage Policy

TrackMind operates under an explicit data ownership principle: all documents uploaded by a client remain the exclusive property of that client. TrackMind commits to three non-negotiable data usage restrictions:

- Client documents are never used to train, fine-tune, or otherwise update the underlying LLM. The Claude Sonnet model used for inference operates under Anthropic's API terms, which explicitly prohibit training on API inputs.
- Client documents are never shared with, or made accessible to, any other client. The multi-tenant isolation is enforced at the architecture level, not just at the policy level.
- Client documents are deleted from TrackMind's infrastructure upon contract termination, within a defined retention window specified in the service agreement.

9.3 The Data Flywheel — What It Is and What It Is Not

TrackMind's competitive moat includes a data flywheel: the system improves over time as more compliance queries are processed and approved by human reviewers. The flywheel does NOT involve client documents. It involves anonymised resolution metadata: the fact that a certain type of TSI/NTR conflict was resolved in a certain way, stripped of any client-identifying information. Concretely: if a client's engineer approves a response resolving a conflict between LOC&PAS TSI Article 4.2.3.2 and the French NF F31-054 standard, TrackMind records that this conflict type was resolved successfully. It does not record the client's vehicle specifications, the exact wording of their technical file, or any other proprietary content.

9.4 Data Protection as a Competitive Differentiator

Property	General-purpose LLM (Claude Projects, ChatGPT)	TrackMind
Document storage location	Provider infrastructure (Anthropic/OpenAI servers). Client has no control over data residency.	TrackMind-controlled GCP infrastructure. Documents never transmitted to third-party vector DB providers.

Property	General-purpose LLM (Claude Projects, ChatGPT)	TrackMind
Multi-tenant isolation	Not applicable — single user session, no persistent multi-tenant architecture.	Architecture-level isolation. Separate ChromaDB collection per client. Cross-client retrieval is structurally impossible.
Used for model training	Depends on provider terms. Consumer tiers often train on user inputs by default.	Never. Contractual prohibition. Anthropic API terms explicitly prohibit training on API inputs.
Audit trail	No structured audit log. Chat history only, not suitable for regulatory documentation.	Full audit log: timestamp, assessor ID, query, AI response, human decision. Retained for regulatory compliance.
Suitability for safety-critical regulated use	Not designed for this context. No confidence scoring, no human-in-the-loop, no document provenance.	Designed for this context. Confidence scoring, mandatory human approval, full source citations, complete audit trail.

10. Regulatory Corpus Maintenance Strategy

TrackMind is only as reliable as the regulatory documents it indexes. A RAG system that retrieves outdated normative text produces outputs that are not just unhelpful — they are potentially dangerous in a safety-critical certification context. Maintaining a current, accurate corpus of TSI documents and National Technical Rules is therefore not an operational afterthought: it is a core product requirement and a meaningful competitive moat against general-purpose LLM alternatives.

Source type	Examples	Update frequency	Management complexity
European TSIs	LOC&PAS TSI, ENE TSI, CCS TSI, INF TSI. Published in the EU Official Journal.	Every 3-5 years. ERA notifies all revisions. Highly predictable.	Low. Easy to monitor, high impact per update.
National Technical Rules (NTR)	French Arrete, Spanish RD, German EBO, Italian PGOS. Notified via ERA ERADIS database in 24 languages.	Irregular. ~5-10 significant changes per year across key countries.	Medium. Requires automated monitoring + human review per update.
NB-Rail Recommendations for Use	Interpretive guidance documents published by the NB-Rail coordination group.	Infrequent but high-value. Published on the NB-Rail website when new interpretations are agreed.	Low. Periodic manual check sufficient.

10.1 The Three-Layer Maintenance Strategy

Level 1 — Automated source monitoring

ERA publishes notifications for all TSI revisions and NTR changes through the ERADIS database and official channels. TrackMind implements a lightweight automated monitoring service — a scheduled job running weekly — that checks these sources for new or updated documents. Setup cost: 1-2 days of engineering work once. Ongoing cost: negligible (automated).

Level 2 — Human review and re-ingestion

When the monitoring service flags an update, the full-stack AI engineer reviews the changed document, confirms it supersedes a previous version, re-ingests it into ChromaDB, and deprecates the old chunks. Based on historical ERA data, approximately 5-10 significant NTR changes per year affect the key countries relevant to TrackMind's clients. Each re-ingestion requires approximately 1-2 hours of engineering time. Total annual overhead: 10-20 hours — less than 1% of the engineer's working time.

Level 3 — Automatic chunk invalidation

Every document chunk indexed in ChromaDB carries metadata tags including the source document version, publication date, and supersession status. When a document is updated, chunks from the previous version are automatically marked as deprecated before the new version is re-ingested. This prevents the system from serving outputs based on obsolete regulatory text during the transition period.

10.2 Regulatory Currency as a Competitive Differentiator

A general-purpose LLM has a training data cutoff. Its knowledge of EU railway regulations is frozen at that date. When the French transport authority revises an NTR, when ERA publishes an updated LOC&PAS TSI application guide, or when a new NB-Rail Recommendation for Use clarifies an ambiguous compliance case, the general-purpose LLM remains unaware.

TrackMind operates on a fundamentally different model: the regulatory corpus is a live, maintained asset, not a static training snapshot. Every response cites a specific document version with a publication date. The client can verify at any moment that the normative text underlying a compliance output is the current applicable version.

Corpus SLA — a contractual commitment no LLM can make

TrackMind can offer clients a contractual Service Level Agreement on corpus currency: NTR changes notified on ERA ERADIS are integrated into the corpus within 30 days of publication. TSI revisions are integrated within 60 days of entry into force. This commitment is verifiable and enforceable in a service contract. No general-purpose LLM provider offers — or could offer — an equivalent guarantee. The corpus SLA transforms regulatory currency from a product feature into a contractual obligation that increases switching costs and deepens client trust.

11. Regulatory Compliance: GDPR and EU AI Act

TrackMind operates at the intersection of two major European regulatory frameworks: the General Data Protection Regulation (GDPR) and the EU Artificial Intelligence Act (AI Act). This section explains TrackMind's position under both frameworks, and demonstrates that the product is designed to be compliant by architecture rather than by retrofit.

11.1 GDPR

The documents processed by TrackMind consist almost entirely of technical, non-personal data: engineering specifications, dimensional requirements, test protocols, and regulatory articles. Personal data, where present, is limited to names and signatures of engineers who authored or approved sections of technical documents.

Under the GDPR, TrackMind acts as a data processor on behalf of the manufacturer client, who is the data controller. The manufacturer determines the purpose and means of processing; TrackMind processes the data solely to deliver the contracted service. This relationship is governed by a standard Data Processing Agreement (DPA) included in every service contract.

11.2 EU AI Act — Risk Classification

Why high-risk does not apply

The EU AI Act classifies as high-risk AI systems that manage or control safety-critical transport infrastructure — for example, systems that manage railway signalling, control train movements, or operate safety components in rail networks. TrackMind does none of these things. It does not control any physical infrastructure, does not make operational decisions about train movements, and has no interface with safety-critical systems.

Furthermore, the AI Act explicitly exempts from high-risk classification systems that perform a preparatory task to an assessment and are not designed to replace the previously completed human assessment without proper human review. TrackMind is precisely this: a preparatory tool that retrieves regulatory text and drafts a structured analysis for a human engineer to review, correct, and approve.

TrackMind is classified as Limited Risk

TrackMind falls under the Limited Risk category. The primary obligation for limited-risk AI systems is transparency: users must be informed that they are interacting with an AI system. TrackMind satisfies this by design — every output is explicitly labelled as AI-generated, includes a confidence score, and requires explicit human approval before it can be used in any official documentation.

11.3 Safety Measures Beyond Compliance

Although TrackMind is classified as limited-risk and therefore subject only to transparency obligations, the product implements safety measures that go significantly beyond what the AI Act requires.

Safety measure	How it works	Why it matters
Human-in-the-loop (mandatory)	Every AI-generated response must be explicitly approved, edited, or rejected by a qualified engineer before it can be used. No output is ever automatically incorporated into a technical file.	Eliminates automated decision-making in a safety-critical context. The engineer retains full professional responsibility.
Confidence scoring (Green / Amber / Red)	Green (>90%): direct article citation found. Amber (70-90%): relevant articles found, interpretation required. Red	The system refuses to hallucinate when it lacks sufficient evidence. Engineers are never misled by false confidence.

Safety measure	How it works	Why it matters
	(<70%): document gap detected, system blocks approval.	
Full audit log	Every query, AI response, and human decision is logged with timestamp and assessor ID. The log is immutable and retained for the duration of the service contract.	Provides full traceability for regulatory inspections. Every compliance decision can be traced back to the specific document version and engineer who approved it.
Source citations with document version	Every response includes explicit citations to the specific TSI article, NTR section, and document version on which it is based.	Prevents over-reliance on AI output. The engineer is always one click away from the primary source.
AI transparency labelling	Every output is explicitly labelled as AI-generated. Users are never in doubt about the nature of what they are reading.	Satisfies the primary obligation of the AI Act for limited-risk systems.

TrackMind is compliant with the AI Act by design, not by retrofit. The human-in-the-loop architecture, confidence scoring, audit logging, and source transparency were built into the product from day one because they make the product more useful and trustworthy in a professional context — not because a regulation required them.

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